

Summary of Initial Insights:

The analysis of the loan data has yielded crucial insights that can significantly impact the bank's decision-making process regarding loan approvals. Key observations include:

1. Applicant Income:

- The average income of loan applicants is around \$3,605, with a wide range from \$150 to \$9,703. The variation in income highlights the diverse financial backgrounds of applicants.

2. Co-applicant Income:

- The average co-applicant income is approximately \$1,263. A notable outlier with a co-applicant income exceeding \$30,000 suggests the need for further investigation.

3. Loan Amount:

- The average loan amount requested is about \$105, with a range from \$9 to \$150. Loan amounts vary widely, indicating diverse financial needs among applicants.

4. Credit History:

- A significant number of applicants have a good credit history (1.0), positively influencing loan approval.

5. Gender and Marital Status:

- Males and married individuals dominate the pool of loan applicants. These demographics may be considered when tailoring marketing strategies.

Important Pattern:

- People with poor credit history (0.0) along with co-applicant income (0) applications are completely rejected irrespective of their income, loan amount and education. However, a person who got the loan which is an exception and might be a special case or anomaly or fraudulent activity which needs to be investigated.

- Also, bank has been lenient towards people who are self-employed, having 3+ dependents, whose income is almost double than the average income and have good credit history (1.0) compared to people who have same similarities except for not self-employed and 0 dependents.

- Credit history is the only independent variable which is majorly deciding loan approval status. Applicant income, co-applicant income, loan amount are supporting factors to decide loan status.

Recommendations:

- Bank needs to approve loans for people who have good credit history and as well as good income to increase their business.

- Based on only credit history, it has not given loans to many people, it must consider other factors like income of the person, dependents, current living area and can grant loans which will build a strong user base.